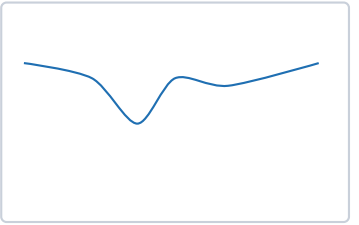
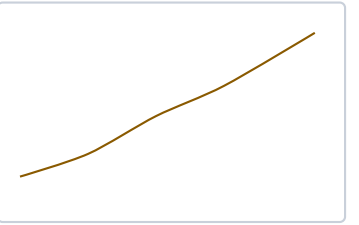


Ultracool Y / T Brown Dwarf



Icy KBO / M-Dwarf Giant



Dusty T Tauri Star

Gating Contract (`step5_spectrum.py` and `app.py`): Standard 2-to-6 band cutouts yield 12–18 observed SED flux points. The three-index categorical classifier explicitly requires ≥ 30 spectral channels before reporting a physical label, preventing hallucinations from sparse cutouts.